

Question:

If the sum of all the angles except one of a convex polygon is \$2190^{\circ}\$, then the number of sides of the polygon must be

(A) 13 (B) 15 (C) 17 (D) 19 (E) 21

Reasoning + Answer:

<think>

1. The sum of all interior angles of a polygon with \$n\$ sides is given by the formula :

$$\text{Sum of all interior angles} = 180(n - 2)^{\circ}$$

2. We know that the sum of all angles except one angle is \$2190^{\circ}\$. Let \$x\$ be the excluded angle . Therefore , we can write :

$$180(n - 2) = 2190 + x$$

3. Since \$x\$ must be an interior angle of a convex polygon , it can be at most \$180^{\circ}\$. Hence , \$x = 180^{\circ}\$.

4. Substitute \$x = 180^{\circ}\$ into the equation :

$$180(n - 2) = 2190 + 180$$

5. Simplify the right-hand side :

$$180(n - 2) = 2370$$

6. Divide both sides by 180 to solve for \$n\$:

$$n - 2 = \frac{2370}{180}$$

7. Calculate the division :

$$n - 2 = 13.1667 \quad (\text{approximately})$$

8. Add 2 to both sides to solve for \$n\$:

$$n = 15$$

9. Therefore , the number of sides of the polygon is 15 .

</think>

<answer>

15

</answer>